Chandan Kumar

Rust Backend Developer

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EDUCATION

Punjab Engineering College (PEC) — Chandigarh, India
Bachelor of Technology in Electrical Engineering | Dec 2020 – Jun 2024 | CGPA: 7.08
Relevant courses: Data Structures, Algorithms, Database Management Systems, Operating Systems

EXPERIENCE

Rust Developer — Webmob Software Solutions → Webcom Systems
2023 – Present

- Designing and developing scalable backend services in Rust for production-grade REST APIs and microservices
- Building asynchronous API handlers and background tasks using Actix-Web and Tokio for high-performance workloads
- Integrating PostgreSQL and MongoDB with optimized queries, indexing, and efficient data modeling
- Implementing authentication, authorization, request validation, and RBAC for secure API access
- Collaborating with cross-functional teams on development, debugging, performance improvements, and code reviews
- Managing Git workflows, branching strategies, CI pipelines, and deployments using GitHub

PROJECTS

XchangeOn – Matching Engine
Rust, NATS JetStream, Kafka, Redis, Hexagonal Architecture

- Designed and built a native Rust futures matching engine microservice (hexagonal architecture, NATS JetStream) supporting leveraged perpetual futures – order matching, leverage/margin validation, TP/SL passthrough, cancel/replace – alongside a production spot exchange, with zero regressions across 75+ automated tests
- Built a native algorithmic market-maker engine from scratch – mark-price EMA blending, volatility-adaptive price windows, and three liquidity strategies – replacing a legacy Node.js bot loop with a lower-latency, in-process design
- Implemented a dual-write audit pipeline (Redis pub/sub for live delivery, Kafka for durable audit trail) with code-managed topic retention, tuned for sub-3-second live market data delivery
- Diagnosed and resolved a live production incident spanning SASL/TLS Kafka authentication and custom CA trust across a multi-stage CI/CD pipeline, restoring data flow across 6 Kafka topics without downtime
- Verified system behavior end-to-end against real external market data (live Binance feeds) across NATS, Redis, and Kafka before deployment

G20 Multi-Exchange Trading
Rust, WebSockets, REST APIs, Tokio, HMAC

- Architected and implemented a high-performance exchange connector framework in Rust for G20 Rapid Engine, supporting 50+ cryptocurrency exchanges including Binance, Coinbase, Kraken, OKX, Bybit, and LMAX
- Built unified REST and WebSocket interfaces for order placement, cancellation, trade execution, and real-time market data streaming across spot, futures, and DEX markets
- Developed low-latency WebSocket streaming infrastructure using Tokio with heartbeat management, auto-reconnection, and message queuing for real-time balances, order state changes, and fills
- Implemented secure authentication mechanisms including Bearer tokens, OAuth 2.0, and HMAC-SHA256 signature generation with automatic token refresh and credential isolation
- Designed standardized internal data models (ExchOrder, ExchBalance, ExchFillDetail) to normalize heterogeneous exchange APIs and enable consistent trading logic

LoveToPlay.Golf (LTPGolf)
Rust, Actix-Web, PostgreSQL, JWT, FCM, Docker

- Built backend with Rust + Actix-Web powering competitions, scoring, handicap engine, chat, and subscriptions
- Designed PostgreSQL schemas and implemented 100+ APIs with JWT authentication, RBAC, and validation layers
- Implemented real-time chat, FCM push notifications, presence, polls, and timed voting flows
- Added cron-style scheduled jobs for reminders, billing, leaderboard refresh, and event automation
- Optimized PostgreSQL queries using indexes and CTEs, significantly improving latency for leaderboard and scoring endpoints

M24 Sniper Crypto Trading Bot
Rust, Crypto APIs, WebSockets

- Developed a crypto trading bot to detect rug-pull scams and high-risk patterns in newly launched tokens
- Implemented real-time transaction monitoring, alerting, and automated trade execution using exchange WebSockets
- Designed concurrency-safe order execution using Rust ownership and async primitives, preventing race conditions
- Built latency-sensitive pipelines for price-feed and mempool monitoring to react quickly to market moves
- Integrated with multiple exchange APIs for real-time data ingestion and order execution

SKILLS

Languages: Rust, C++, JavaScript, TypeScript, SQL, Python
Frameworks & Libraries: Actix-Web, Axum, Tokio, Serde, Diesel, SQLx
Databases & Messaging: PostgreSQL, MongoDB, Supabase, Redis, NATS JetStream, Kafka
Cloud & DevOps: AWS S3, Docker, GitHub Actions, Firebase FCM
Tools: VS-Code, Git, GitHub, Postman, Linux
Concepts: Data Structures, Algorithms, OOP, Microservices, REST APIs, Async Programming